

ABER: Supplementary Material

Anonymous submission

Scope and Method Identity

This supplement concerns only ABER, the forward/turn sampled-data filter defined in the main paper. Its deployed action is the normalized pair (u^f, u^w) ; it does not assume a Cartesian acceleration command. The implementation entry point is `safe_rl/filters/aber.py`, which exports `ABERFilter` from `safe_rl/filters/brake_manifold.py`. The latter filename and the `BrakeManifoldFilter` alias are retained only so historical configuration files and checkpoints remain loadable.

Earlier diagnostic material in the repository studied other safety-filter designs. Their propositions and experiments are not used as evidence for the current paper. The current empirical claims are backed by eight checked-in evidence groups:

1. `data/aber_property_tests.json`, the seeded falsification audit;
2. `data/certified_diffdrive_benchmark.json`, generated by the fixed theorem-envelope benchmark;
3. `data/aber_component_ablation.json` and its raw JSONL.GZ, covering 295,200 component-study records;
4. `data/safetygym_aber_matrix.json` and the exactly reproduced `data/safetygym_aber_matrix_retrained.json`, audited aggregates of 30 training cells and 1,500 deterministic evaluation episodes;
5. `data/aber_runtime_scaling.json` and its raw JSONL.GZ, covering one million timed projection calls;
6. `data/aber_paired_matrix.json`, the strict-ABER paired audit and its independent verification report;
7. `data/aber_safetygym_contract_v1.json` plus 30 raw trace files, the frozen 600-episode A1–A2 substep audit reported in Section ; and
8. `full_30_checkpoint_aggregate.json` and `cross_task_audit.json`, the complete filter-off/strict/ABER-LR 30-checkpoint analysis.

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The paired evidence additionally hashes all 30 exact policy checkpoints, resolved training configurations, raw episode and step files, and generating scripts. These checkpoints are not used by any theorem or component-study claim.

Acceptance Envelope and Recovery

For obstacle i , let $d_i(q) = \|q - o_i\| - r_i$ and $D(s) = s^2/(2a_b)$. ABER certifies the intersection

$$\mathcal{C} = \{(q, s) : d_i(q) - D(s) \geq 0 \text{ for every } i\}.$$

The normal branch uses the measurable envelope

$$s^+ \leq \max\{s, v_{\max}|u^f|\}, \quad \ell \leq \Delta t \max\{s, v_{\max}|u^f|\}.$$

Writing z for the common upper bound, the sufficient one-step condition is

$$\Delta t z + D(z) \leq d_{\min}, \quad d_{\min} = \min_i d_i(q).$$

Its nonnegative inverse is

$$z_{\max} = \min\left\{v_{\max}, -a_b \Delta t + \sqrt{(a_b \Delta t)^2 + 2a_b d_{\min}}\right\}.$$

If $s \leq z_{\max}$, the implementation clips only u^f to $[-z_{\max}/v_{\max}, z_{\max}/v_{\max}]$ and retains the requested turn. Otherwise it sends the executable recovery $(0, 0)$.

Cap maximality and its boundary

For $d \geq 0$, define $g(z) = \Delta t z + z^2/(2a_b)$. Since $g'(z) = \Delta t + z/a_b > 0$ for $z \geq 0$, g is strictly increasing and the unclipped expression above is its unique nonnegative inverse. Consequently,

$$z_{\max}(d) = \max\{z \in [0, v_{\max}] : g(z) \leq d\}.$$

When $z_{\max} < v_{\max}$, every larger speed violates the robust one-step condition. This is a sharpness statement only within ABER’s direction-independent A1 envelope: a controller using obstacle bearing or turn geometry may certify a larger action in a particular state.

Semi-implicit recovery check

The main theorem accepts any measured recovery envelope satisfying

$$\ell^{\text{rec}} + D(s^+) \leq D(s). \quad (\text{S1})$$

For the benchmark plant,

$$s^+ = \max\{s - a_b \Delta t, 0\}, \quad \ell^{\text{rec}} \leq \Delta t s^+.$$

When $s < a_b \Delta t$, both s^+ and the upper bound on path length are zero. Otherwise,

$$\begin{aligned} \ell^{\text{rec}} + D(s^+) &\leq \Delta t(s - a_b \Delta t) + \frac{(s - a_b \Delta t)^2}{2a_b} \\ &= \frac{s^2}{2a_b} - \frac{1}{2}a_b \Delta t^2 \leq D(s), \end{aligned}$$

which proves (S1). This derivation is specific to the accepted low-level envelope; it is not inferred merely from the commanded setpoint.

Why one recovery covers multiple obstacles

Distance to each static obstacle is 1-Lipschitz with respect to path length: $d_i(q^+) \geq d_i(q) - \ell$. Therefore, in the normal branch,

$$d_i(q^+) - D(s^+) \geq d_{\min} - \Delta t z_{\max} - D(z_{\max}) \geq 0$$

for every i . In recovery, (S1) gives

$$d_i(q^+) - D(s^+) \geq d_i(q) - D(s) \geq 0$$

for every i simultaneously. No intersection of obstacle-dependent directional half-spaces is required. Heterogeneous physical radii are supported by `reset(obstacles, obstacle_radii=...)`; callers that omit the optional list retain the configured common hazard radius. Position uncertainty is added to every keepout radius before clearances are computed.

Code-to-Paper Mapping

The public filter rejects `action_form=cartesian`. Historical safety identifiers `brake_manifold` and `saber` dispatch to the same `ABERFilter`; they do not select separate algorithms. The focused test command is:

```
PYTHONPATH=. pytest -q tests/test_
brake_manifold_sampled_data.py
PYTHONPATH=. pytest -q tests/test_
aber_component_ablation.py
```

The complete focused filter suite reports 25 passed tests in the archived environment.

Falsifiable Property Audit

The seeded script `experiments/aaai/certify_aber_properties.py` records source hashes, parameters, environment versions, and up to 20 counterexamples per failed property before exiting nonzero. It checks 5000 analytic cap inversions and 26,354 normal/recovery projections over one, two, and eight heterogeneous static obstacles. The latter count includes every normal sample plus every sample for which a certified recovery state exists. No required property fails.

The same script deliberately violates the calibration premise by setting actual braking to $0.5a_b$ while retaining the accepted envelope in the certificate. All 5000 samples then produce a negative next-step barrier. These expected counterexamples demonstrate that $(0,0)$ is a certified recovery only after A2 has been established for the deployed low-level controller.

Paper object	Implementation	Test or data evidence
$d_{\min}$ and $D(s)$	<code>ABERFilter.project</code>	heterogeneous-radius and uncertainty tests
$z_{\max}$ inverse	<code>ABERFilter.speed_cap</code>	5000 inversion/sharpness checks
$(0,0)$ recovery	<code>project; recovery_speed_bound</code>	executable-recovery and adversarial-start tests
forward/turn action map	<code>project</code>	Cartesian action rejection; turn retention
component necessity	<code>experiments/aaai/aber_component_ablation.py</code>	295,200 raw boundary/navigation records
certified benchmark	<code>experiments/aaai/certified_diffdrive_benchmark.py</code>	200 episodes per method across two scene families
Safety-Gym aggregation	<code>experiments/aaai/aggregate_aber_matrix.py</code>	30 manifests, saved Hydra configs, 1500 evaluations
runtime scaling	<code>experiments/aaai/aber_runtime_scaling.py</code>	1,000,000 project timings
paired policy matrix	<code>run_aber_latched_restore_cross_task_multiseed.py; audit_aber_latched_restore_cross_task_multiseed.py</code>	30 checkpoints, 15,000 filtered paired rollouts

Table 1: Direct mapping from the current ABER paper to code and evidence.

Component Necessity and Calibration Study

The component study crosses four control periods $\{0.02, 0.05, 0.10, 0.20\}$ s, four actual/configured braking ratios $\{0.5, 0.75, 1.0, 1.25\}$, and $m \in \{1, 4, 8\}$ obstacles. Its three variants are full ABER, ABER without the sampled displacement $\Delta t z$, and ABER without the distinct $(0,0)$ recovery branch. Each grid cell has 1,000 samples for each boundary challenge and 50 navigation episodes.

Boundary controls

The sampled-term challenge initializes $d = \Delta t s + D(s)$; the recovery challenge initializes $d = D(s)$. The former makes the full normal inequality tight, while the latter requires the A2 witness. Table 2 pools equal-size calibrated or under-braking cells only after all 48 predeclared per-grid checks pass.

One-step certificate violation is deliberately the primary endpoint. A sample can leave the recursively certified set before it crosses the obstacle surface, so zero immediate

Calibration	Variant/challenge	Trials	Viol.
$\rho \geq 1$	Full / sampled term	24,000	0
$\rho \geq 1$	Full / recovery	24,000	0
$\rho \geq 1$	No sampled term / sampled term	24,000	24,000
$\rho \geq 1$	No recovery / recovery	24,000	24,000
$\rho < 1$	Full / recovery	24,000	24,000

Table 2: Predeclared one-step controls. Violation means a negative next-state braking-envelope certificate; no row is selected post hoc.

collisions would not rescue a failed certificate.

Navigation results

Among calibrated episodes, full ABER has 0/1,200 violation episodes and 0/1,200 collisions, with 1,186 successes and 14 horizon timeouts. Removing the sampled term yields 1,200/1,200 violation episodes, 481 collisions, and 719 successes. Removing recovery yields 1,002 violation episodes, 298 collisions, and 902 successes. Under braking ratios below one, full ABER itself produces 873/1,200 violation episodes and 539 collisions, which is the intended negative control for a false A2 calibration.

The artifact stores 288,000 individual boundary records and 7,200 navigation records as deterministic gzip JSON Lines. Two complete runs produced the same raw SHA-256, recorded in the manifest. The aggregate also records hashes of the exact filter and generation script.

Certified-Envelope Benchmark Details

The benchmark uses $\Delta t = 0.1$ s, $v_{\max} = 1$ m/s, $a_{\text{acc}} = a_b = 0.9$ m/s², $|\omega| \leq 1.5$ rad/s, and $r_i = 0.25$ m. The state update is semi-implicit: the bounded velocity servo is applied first, then the next speed is held over the position update. All methods use the same reactive navigator and the same 100 fixed seeds in each scene. The three comparison filters are fixed geometry-specialized reference implementations, not claimed reproductions of every tuning choice in their source papers.

Scene	Method	Coll. (%)	Succ. (%)	Time (s)	min clr. worst (m)	Path (m)	Interv. mean
Single	Braking CBF	0	100	5.80	0.290	4.30	40.85
	SD-CBF/ZOCBF	0	100	4.90	0.185	4.30	33.77
	Collision cone	0	100	5.91	0.083	4.41	59.10
	ABER	0	100	6.46	0.016	4.22	32.26
Multi	Braking CBF	0	100	7.22	0.220	4.85	70.00
	SD-CBF/ZOCBF	0	100	5.55	0.046	4.86	55.39
	Collision cone	0	100	7.33	0.001	5.06	73.28
	ABER	0	100	7.22	0.016	4.80	60.40

Table 3: Full fixed-envelope aggregate. Each cell has 100 episodes.

ABER invokes recovery 866 times in the single-obstacle set and 1335 times in the multi-obstacle set, with zero recorded certificate violations. All four methods have zero

collisions and 100% goal success. Accordingly, this experiment supports implementation consistency and task completion inside the accepted envelope; it does not support an unsafe-baseline claim.

Safety-Gymnasium Audit

The matrix crosses three tasks, two reward modes, and five training seeds. Each run uses 200K environment steps and 50 deterministic evaluation episodes. All 30 cells disable EKF/IMU estimation and action-correction calibration. The saved Hydra configurations are checked for a normalized forward/turn action, $a_b = 0.9$, $\Delta t = 0.02$, $v_{\max} = 1.5$, a 0.05 m safety margin, and task-specific hazard radius. The aggregation script aborts on a missing run, missing metric, unexpected configuration value, or an evaluation count other than 50.

Mode	Task	Cost	Reward	Succ. (%)	worst clr.	deep steps
Native	Goal1	79.24 $\pm$ 93.52	-6.97 $\pm$ 9.78	39.2 $\pm$ 18.1	-0.075	26.30
Native	Push1	5.85 $\pm$ 7.91	0.28 $\pm$ 0.14	1.6 $\pm$ 1.7	-0.497	58.29
Native	Goal2	26.73 $\pm$ 40.55	-1.33 $\pm$ 3.79	21.6 $\pm$ 11.3	-0.039	0.00
+sim-BRT	Goal1	16.86 $\pm$ 20.35	-1.53 $\pm$ 2.22	24.8 $\pm$ 17.8	-0.017	0.00
+sim-BRT	Push1	36.48 $\pm$ 52.32	0.23 $\pm$ 0.05	0.8 $\pm$ 1.8	-0.468	42.03
+sim-BRT	Goal2	21.44 $\pm$ 14.40	-0.70 $\pm$ 0.82	12.0 $\pm$ 11.3	-0.009	0.00

Table 4: Audited ABER matrix. Mean $\pm$ sample standard deviation is across five seeds. Worst clearance is over 250 evaluation episodes per cell; deep steps use the pre-declared -0.05 m threshold.

These results are empirical and are not presented as validation of the global A1-A2 servo envelope. sim-BRT reduces cost on Goal1 and Goal2, increases it on Push1, and lowers mean goal success in all three tasks. Push clearance includes box-hazard geometry, which lies outside the robot-obstacle certificate. The paper therefore reports native and shaped training separately and makes no claim that reward shaping is uniformly beneficial.

Runtime Scaling Audit

The runtime experiment invokes the submitted `ABERFilter.project` implementation directly. It crosses $m \in \{0, 1, 2, 4, 8, 16, 32, 64, 128, 256\}$, with 20 repeats and 5,000 calls per repeat, for exactly one million retained timing records. Positions, velocities, and actions are generated before timing; only `project` is inside `perf_counter_ns`. Serialization, policy inference, and environment stepping are excluded. Count order is reversed on alternating repeats, each cell receives 200 warm-up calls, and the process is pinned to one core of an Intel i7-13850HX.

For positive counts, least-squares fitting of the pooled per-count median gives $0.3168m + 17.1084 \mu\text{s}$ with $R^2 = 0.99961$. This is empirical scaling on one machine, not the complexity proof; the $O(m)$ result follows independently because the implementation performs one vectorized distance reduction and a constant amount of scalar work. Exact source and raw-data hashes are recorded in the aggregate and checksum manifest.

m	Median (μ s)	p95 (μ s)	p95/20 ms (%)
0	6.14	6.46	0.032
1	17.96	18.82	0.094
8	19.35	21.57	0.108
32	27.80	30.80	0.154
128	57.35	61.79	0.309
256	98.53	103.76	0.519

Table 5: Observed wall time over 100,000 calls per listed count except that all ten counts, including omitted intermediate rows, are present in raw data.

Paired Learned-Policy Safety–Liveness Audit

We retrain 30 policies: native and sim-BRT training on Goal1, Push1, and Goal2, with five training seeds per cell and 200K environment steps per policy. Each atomic checkpoint contains the actor tensors, resolved configuration, training metadata, and SHA-256 digest. The resulting training-matrix aggregate exactly reproduces all 192 scalar statistics in the earlier audited aggregate (maximum absolute difference zero).

Each checkpoint is evaluated with deterministic policy actions under three execution conditions: filter off, strict ABER, and the frozen ABER-LR extension. Each filtered variant is paired with filter-off execution through the same checkpoint, reset seed, and canonical initial snapshot, including robot, goals, obstacles, walls, and the Push box. Native Goal1 uses reset seeds 44000–44249; the remaining 25 checkpoints use 45000–45249. The two filter-off reruns are identical at the decoded-record level. All checkpoint and initial-state hashes match. No seed is searched, discarded, selectively rerun, or imputed.

ABER-LR uses

$$(\epsilon, a_b, s_{\text{stop}}, u_{\text{restore}}^f, k_\omega) = (0.10, 0.9, 0.05, 0.25, 1.0).$$

Directional restore is enabled and latched; the separate escape heuristic is disabled. These values were frozen before the five remaining mode–task groups were evaluated. Goal1 was the development group used to choose the candidate; the cross-task matrix is therefore the transfer test. We report all groups rather than selecting only those that pass the development gate.

For each condition, the primary joint endpoint is

$$\text{SCR} = \Pr(\text{success} \wedge \neg \text{geometric collision}).$$

Collision is negative physical surface clearance; Push uses the minimum of robot–hazard and box–hazard clearance. SCR is always accompanied by raw success, collision, unsafe success, and timeout, so it does not conceal the marginal tradeoff or introduce a tuned scalarization weight. Each checkpoint contributes one 250-episode rate difference. Reported 95% intervals are percentile intervals from 50,000 seeded bootstrap resamples of the 30 checkpoint differences; episodes are never treated as independent training replicates. Equal-group and equal-checkpoint weighting coincide because every mode–task group contains five checkpoints.

Failure Definition and Rejected Repairs

A negative test is not automatically a collision failure. We distinguish: (i) *safety failure*, crossing the frozen collision bound; (ii) *liveness failure*, missing the success or recovery-tail target; (iii) *coordination failure*, missing the joint SCR target despite an improved marginal; and (iv) *transfer failure*, losing the required direction or magnitude across training seeds or tasks. A separate *contract failure* means that A1–A2 cannot be attached to the empirical plant; it does not erase an empirical collision result.

The 250-pair yaw-escape pilot required at least +10 success points with no more than +3 collision points. The later cross-task gate requires at least +10 success points, no collision increase, at least +5 SCR points, and SCR improvement in at least four of five training seeds. The 50-pair margin screen was discovery-only: it selected a candidate but could not pass the method by itself. Table 6 retains the negative results that determined the final mechanism.

The structural difficulty is therefore two-sided. With zero translational authority, recovery cannot increase clearance; changing a_b changes when recovery is entered but not how it exits. Once translation is introduced, the margin must be large enough to absorb its empirical risk but not so large that early intervention recreates the original liveness loss. Latching is needed for persistence until certificate re-entry. These coupled choices explain why a single collision or success number is not an adequate test.

Strict ABER versus filter off. Equal-weight checkpoint means show the intended safety effect and the liveness failure: collision falls from 43.19% to 7.11%, while success falls from 42.48% to 19.92% and timeout rises from 57.52% to 80.08%. SCR falls from 25.96% to 19.88%. Thus collision reduction alone is not an adequate summary of a safety filter attached to a goal-directed policy.

ABER-LR versus strict ABER. ABER-LR raises success by 9.79 points (checkpoint bootstrap 95% CI [6.49, 13.37]), lowers collision by 2.89 points ([1.11, 5.13] reduction), raises SCR by 9.67 points ([6.37, 13.33]), and lowers timeout by 9.79 points ([6.47, 13.40] reduction). All six group means and 28/30 checkpoint cells improve SCR. The strong development-sized gate passes in three groups: Native Goal1, Native Goal2, and sim-BRT Goal1. Native Push1 is the only group with a collision increase (+0.48 points), and its unfiltered success is only 5.12%, limiting the available completion signal.

The three gate failures have different meanings. Native Push1 is a safety and coordination miss (+0.48 collision and only +0.72 SCR points). sim-BRT Push1 moves collision and SCR in the favorable direction but cannot reach the magnitude gate because its unfiltered success is only 4.64%. sim-BRT Goal2 improves SCR in all 5/5 checkpoints but its +8.00-point mean misses the +10 success threshold. The latter two are magnitude/transfer limitations, not reversals of the recovery mechanism.

ABER-LR versus filter off. Collision falls by 38.97 points and unsafe success by 16.36 points. Mean SCR rises by 3.59 points (95% CI [1.03, 6.37]), although raw success falls by

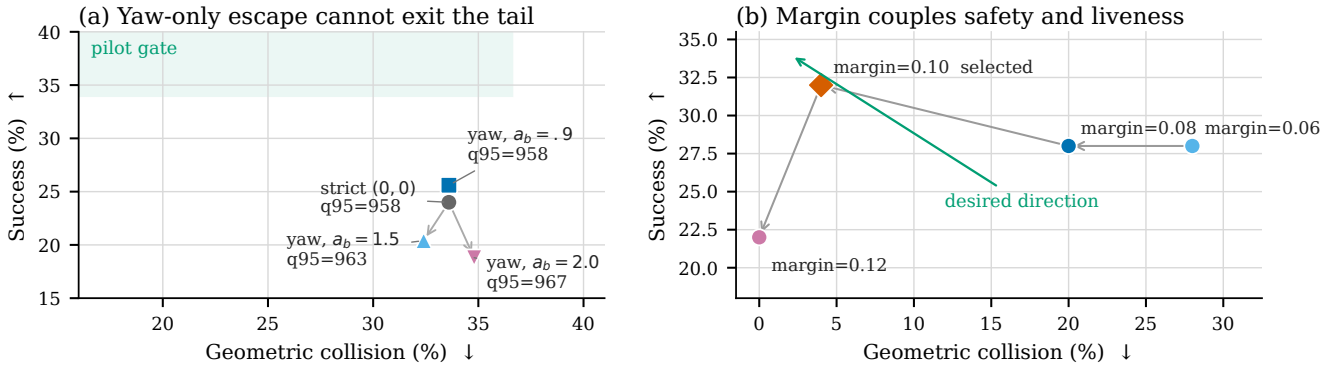


Figure 1: Failure landscape on the Goal1 development checkpoint; upper-left is favorable. (a) Each yaw-only point uses 250 matched episodes. The shaded gate requires success $\geq 34\%$ and collision $\leq 36.6\%$; q95 recovery streaks remain near the 1000-step horizon. (b) A separate 50-pair discovery screen exposes the margin tradeoff. Margin 0.10 was selected for disjoint confirmation; arrows connect settings only to show the frozen search order, not a continuous interpolation.

Attempt (Goal1, one checkpoint)	Pairs	Succ.	Coll.	Rec.	q95	Decision and diagnosed difficulty
% except q95 steps						
Strict (0, 0) recovery	250	24.0	33.6	72.0	958	Liveness failure: the recovery tail occupies nearly the horizon.
Yaw-only escape, $a_b = 0.9$	250	25.6	33.6	70.7	958	Fail: only +1.6 success points; turning does not change position or clearance.
Yaw-only escape, $a_b = 1.5$	250	20.4	32.4	68.8	963	Fail: fewer recovery entries, but the tail persists and success falls.
Yaw-only escape, $a_b = 2.0$	250	18.8	34.8	68.2	967	Fail: success and collision both move unfavorably relative to $a_b = 0.9$.
Latched restore, margin 0.06	50	28.0	28.0	–	–	Discovery-only safety problem: intervention remains too late.
Latched restore, margin 0.10	50	32.0	4.0	–	–	Selected balance; sent to fresh-seed confirmation.
Latched restore, margin 0.12	50	22.0	0.0	–	–	Coordination risk: eliminating collision alone restores conservatism.
Margin-0.10 held-out confirm	250	32.8	7.6	26.8	130	Single-checkpoint pass; still insufficient for cross-task generalization.

Table 6: Rejected-repair audit and the held-out transition to ABER-LR. Success, collision, and recovery occupancy are filter-on rates. Rows use their stated, disjoint discovery/pilot/confirmation schedules and must not be pooled. q95 is the episode-level maximum recovery-streak percentile.

12.77 points and timeout rises by 12.77 points. This SCR improvement is positive in four of six group means and 17/30 checkpoint cells (nonnegative in 19/30), not universal. The supported claim is therefore improved coordination on average, and consistent improvement relative to strict ABER, not dominance of the unfiltered policy.

The full-matrix pattern in Fig. 2 supports the proposed mechanism: strict (0, 0) recovery creates long episodes of immobility that terminate at the horizon, whereas latched directional restore shortens the recovery tail. Timeout falls by the same 9.79 points by which success rises. This is empirical liveness evidence, not a certificate for restore. Push includes movable-box dynamics outside the robot/static-obstacle theorem, so neither Push result is used as theorem validation.

Intervention Event Audit

This separate, earlier strict-ABER audit diagnoses the failure mode that motivated ABER-LR; it is not used to estimate

the three-arm effect sizes in Table 7. We reconstruct events directly from the 11,860,270 paired step records and use only the 6,485,664 filter-on steps. An intervention event, recovery event, or uncertified run is a maximal contiguous run of the corresponding Boolean indicator within one episode. Runs never cross episode boundaries. The six training-mode/task cells retain their original 1,250 filter-on episodes each; no episode, action, or threshold is added by this audit.

Where intervention time comes from. The 6,485,664 filter-on steps contain 480,283 intervention events. Although 60.47% last one step and the median is 0.02 s, the 0.98% lasting at least 100 steps contribute 76.22% of all intervention steps. Every one of the 2,972,091 uncertified steps is also a recovery step; 85.41% of recovery time is uncertified. Thus the primary efficiency bottleneck is entry into, and long dwell within, the uncertified recovery regime, not merely frequent one-step clipping. Debouncing can reduce switching but cannot recover most lost task time without a separately

Training	Task	Off SCR	Strict SCR	LR SCR	$\Delta_{\text{LR-strict}}$ SCR percentage points	$\Delta_{\text{LR-strict}}$ Coll.	$\Delta_{\text{LR-off}}$ SCR	Gate
Native	Goal1	47.52	37.68	56.64	+18.96	-8.80	+9.12	pass
Native	Push1	4.16	3.36	4.08	+0.72	+0.48	-0.08	fail
Native	Goal2	39.60	27.60	46.08	+18.48	-2.80	+6.48	pass
sim-BRT	Goal1	37.68	32.64	43.20	+10.56	-0.72	+5.52	pass
sim-BRT	Push1	2.48	2.16	3.44	+1.28	-4.08	+0.96	fail
sim-BRT	Goal2	24.32	15.84	23.84	+8.00	-1.44	-0.48	fail

Table 7: Complete group-level paired analysis. Negative collision difference is favorable. The strong gate requires at least +10 success points, no collision increase, at least +5 SCR points, and SCR improvement in at least four of five training seeds.

certified directional transition model.

The mutually exclusive step decomposition is 1,753,037 unchanged steps (27.03%), 1,252,911 normal projection steps (19.32%), 507,625 certified recovery steps (7.83%), and 2,972,091 uncertified recovery steps (45.83%). Only 4.30% of intervention events ever touch an uncertified state, yet uncertified states contribute 62.80% of all intervention time. Likewise, 5.02% of uncertified runs last at least 100 steps but contribute 95.62% of uncertified time. These two scales motivate separate remedies: least-change directional certification for ordinary clipping, and prevention of entry into long uncertified recovery for the dominant tail.

The complete machine-readable output is reproduced/aber_intervention_event_audit.json; the corresponding flat cell table is reproduced/aber_intervention_event_audit.csv. The audit is descriptive: it does not choose a post hoc trigger threshold and it does not establish why a particular MuJoCo transition became uncertified.

Held-Out Safety-Gym A1–A2 Substep Audit

We tested whether the MuJoCo interface used by the paired Safety-Gym study accepts the plant contract assumed by the conditional safety theorem. The protocol was frozen before execution (SHA-256 4eaecac9c0c31efcb97fde888fb8594974a21fe54f7f23cd0e5205198061cfe2). It covers both training modes, all three tasks, and all five checkpoints per cell: 30 checkpoint cells, 20 fresh reset seeds per checkpoint, 600 episodes, and 516,966 held actions. The held-out seeds 271000–271019 are disjoint from all strict and ABER-LR paired evaluation schedules. No actor training, model fitting, threshold search, or post-hoc tolerance selection was performed.

For every 20-ms held action, the collector records all ten actual MuJoCo substeps at 2 ms resolution. The frozen primary stratum requires a certified state at the beginning of the hold and no non-ground contact in any substep. An independent raw-trace auditor checks file/checkpoint provenance, action clipping, substep count and timing, support, and three zero-violation gates: the A1 next-speed and held-path bounds and the A2 recovery inequality for the executed (0, 0) action. All integrity and support gates pass, but each contract gate fails (Table 8).

The V1 decision is therefore a *no-go* for connecting the current $a_b = 0.9$, $v_{\max} = 1.5$, $\Delta t = 0.02$ Safety-Gym ex-

Gate	Support	Violations	Rate	Maximum
A1 next speed	231,236	28,041	12.13%	0.10301 m/s
A1 held path	231,236	3,567	1.54%	0.001300 m
A2 recovery	39,290	35,517	90.40%	0.013107 m

Table 8: Frozen primary-stratum result. “Maximum” is the maximum signed contract residual; the frozen numerical tolerance is 10^{-9} .

periments to the unmodified A1–A2 envelope. This is not a failure of the conditional theorem or a retraction of the paired empirical collision comparison. It establishes the sharper boundary: the model-matched benchmark tests the theorem under its assumptions, whereas the Safety-Gym matrix remains empirical and cannot be called theorem validation. In particular, current Safety-Gym data do not certify (0, 0) as the required recovery and do not yet justify MuJoCo ABER-FP directional-tube confirmation.

The complete frozen protocol, aggregate audit, raw gzip traces, collector, substep instrumentation, independent auditor, plotting script, and focused tests are included in the reproducibility materials. A successor contract must be separately named and pre-frozen. It may restrict the accepted operating region or use a conservative measured envelope conditioned on starting speed, executed forward/turn commands, actuator mode, and contact topology; it must not reinterpret V1 or tune a constant to these held-out failures.

Conditional Directional Fast Path

Why the intervention rule must change. The original ABER cap is maximal under the bearing-independent A1 envelope proved in the main paper. Raising that cap by threshold tuning would discard the premise that makes the cap safe. We therefore study ABER-FP, a conditional extension that credits a favorable turn only when a stronger executed-transition certificate is available, and otherwise returns the unchanged ABER action.

Directional transition contract. For candidate action u , let $\Gamma(x, u)$ be a sound tube containing the complete realized position trace over the next hold interval, let $\bar{s}^+(x, u)$ upper-bound the next translational speed, and let $\underline{d}_i^+(x, u)$ lower-bound endpoint clearance to obstacle i . ABER-FP accepts u

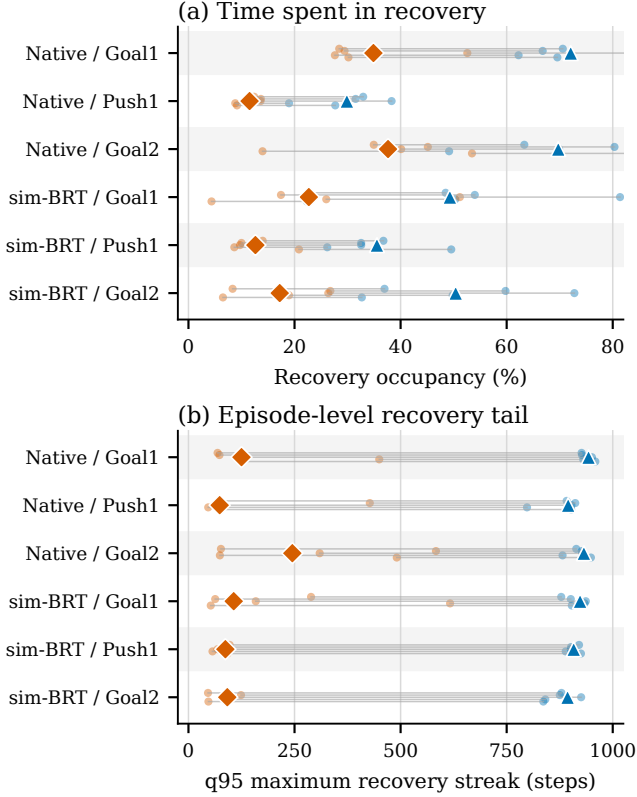


Figure 2: Recovery mechanism by mode-task group. Small points show the five checkpoints; large blue triangles and orange diamonds show strict ABER and ABER-LR group means, with matched checkpoints linked in gray. Mean occupancy falls from 51.15% to 22.76%, and group-average q95 maximum streak from 916 to 121 steps.

only if

$$\min_{y \in \Gamma(x, u)} \|y - o_i\| - r_i \geq 0, \quad (1)$$

$$\underline{d}_i^+(x, u) - D(\bar{s}^+(x, u)) \geq 0 \quad \text{for every } i. \quad (2)$$

Equation (1) prevents intersample keepout entry, while (2) returns the next sample to the original ABER set $\mathcal{C}$.

Complete online algorithm. Let $u_{\text{RL}} = (f, w)$ after range clipping and first compute $u_{\text{base}} = \text{ABER}(x, u_{\text{RL}})$. Form

$$F = f\{1, .75, .5, .25, 0\},$$

$$W = \text{unique}\{\text{clip}(w), \text{clip}(w - .5), \text{clip}(w + .5), -1, 0, 1\}.$$

The candidate library is $Q = \text{unique}((F \times W) \cup \{u_{\text{base}}\})$, hence $|Q| \leq 31$. Sort Q by increasing $\|u - u_{\text{RL}}\|_2^2$. Return the first candidate satisfying (1)–(2) for every obstacle; if none passes, return u_{base} . For m obstacles and the fixed library size, the online work is $O(31m) = O(m)$ and uses no optimization, fitting, interpolation, or actor update.

Safety composition. **Proposition S1.** Suppose the original ABER assumptions A1–A2 hold and the directional contract is sound for every accepted fast-path action. Starting in $\mathcal{C}$,

ABER-FP has the same sampled and intersample safety conclusion as ABER.

Proof. If a fast-path candidate is accepted, (1) keeps its complete held trace outside every keepout and (2) places its endpoint in $\mathcal{C}$. If no candidate is accepted, the output is exactly u_{base} , for which the original ABER theorem preserves the same two properties. Induction over sample intervals proves the claim. $\square$

The proposition is a composition result, not a finite-data guarantee for a learned tube. In the deterministic differential-drive benchmark, $\Gamma(x, u)$ is the exact held line segment and the endpoint is the exact semi-implicit servo transition, so the premise is exact. Safety-Gym and hardware deployment require a separately frozen, held-out accepted path tube; the current 20 braking traces do not validate this directional premise. Accordingly, ABER-FP is not enabled in the MuJoCo claims of the main paper.

The added check raises controller p95 latency from 39.2 to 275.4 μs in the single-obstacle scene and from 36.5 to 345.9 μs in the multi-obstacle scene. The latter is 1.73% of a 20-ms control period. These wall times are machine-specific descriptive measurements; the fixed-library $O(m)$ bound follows from the implementation.

The executable implementation is `experiments/aaai/aber_fastpath_benchmark.py`; tests are in `tests/test_aber_fastpath.py`. Frozen development, held-out, and runtime records are respectively `data/aber_fastpath_benchmark.json`, `data/aber_fastpath_heldout.json`, and `data/aber_fastpath_runtime.json`.

Reproduction and Data Boundary

The local evidence bundle contains a checksum manifest, all 30 evaluation summaries and resolved configurations, all 30 checkpoints, runtime and paired raw records, and a one-command verifier:

```
./verify_artifact.sh
```

Its individual commands are:

```
PYTHONPATH=. pytest -q tests/test_brake_manifold_sampled_data.py
PYTHONPATH=. pytest -q tests/test_aber_fastpath.py
PYTHONPATH=. pytest -q tests/test_aber_component_ablation.py
PYTHONPATH=. pytest -q tests/test_aber_contract_trace.py
PYTHONPATH=. python experiments/aaai/audit_aber_safetygym_contract_v1.py
PYTHONPATH=. python experiments/aaai/certify_aber_properties.py
PYTHONPATH=. python experiments/aaai/certified_diffdrive_benchmark.py
PYTHONPATH=. python experiments/aaai/aber_fastpath_benchmark.py
PYTHONPATH=. python experiments/aaai/aber_fastpath_heldout.py
```

ABER intervention-event audit: 7,500 paired filter-on episodes

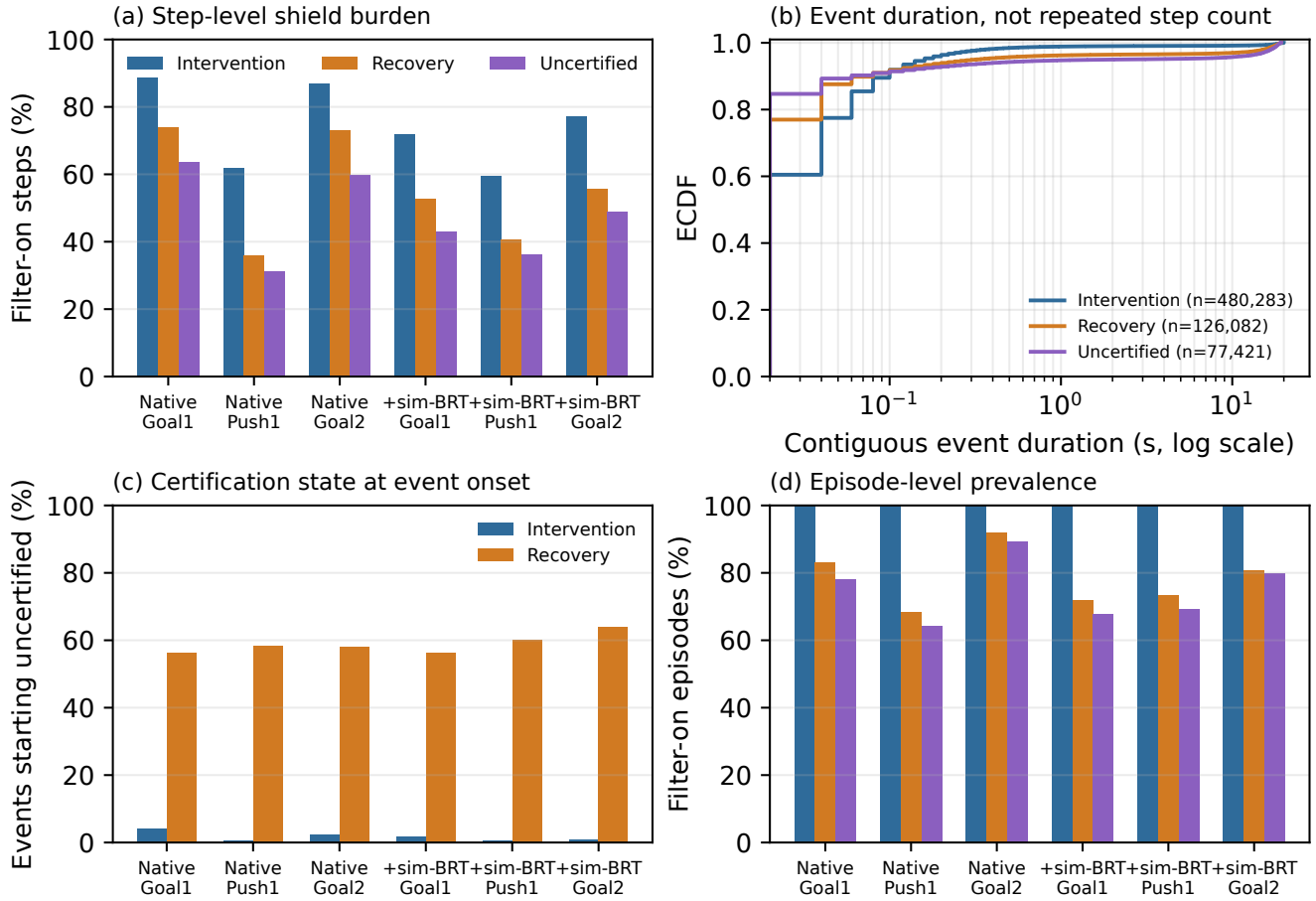


Figure S1: Event-level reconstruction over all 7,500 filter-on episodes. An event is a maximal contiguous run within one episode. Most interventions are one-step corrections, but a small tail of long recovery/uncertified runs dominates time under intervention; overlapping step rates in (a) are therefore not independent categories.

```
PYTHONPATH=. python experiments/aaai/
aber_fastpath_runtime.py
PYTHONPATH=. python experiments/aaai/
aber_component_ablation.py
PYTHONPATH=. python experiments/aaai/
aggregate_aber_matrix.py
PYTHONPATH=. python experiments/aaai/
aggregate_aber_paired_matrix.py
PYTHONPATH=. python experiments/aaai/
analyze_aber_interventions.py
PYTHONPATH=. python experiments/aaai/
verify_aber_paired_matrix.py
PYTHONPATH=. python experiments/
aaai/audit_aber_latched_restore_
cross_task_multiseed.py
PYTHONPATH=. python experiments/aaai/
aggregate_aber_latched_restore_
full_matrix.py
```

```
PYTHONPATH=. python experiments/aaai/
verify_aber_extended_evidence.py
```

The matrix-aggregation command validates existing raw run directories before rewriting the JSON aggregate and main-paper table. It does not synthesize missing cells. If any raw run or saved configuration is absent, it fails rather than imputing a value.

The property, reference, component, and runtime artifacts record Python 3.13.9 and NumPy 2.4.6. The paired rollout used Python 3.10.19, NumPy 1.23.5, PyTorch 2.10.0, and Safety-Gymnasium 1.2.0. The anonymous artifact regenerates the deterministic benchmarks, validates all 30 checkpoint and raw-rollout hashes, and recomputes the full-matrix statistics from frozen audit outputs. Full policy retraining requires the larger external training environment and is outside the minimal artifact. Diagnostic files unrelated to the method and claims in this paper are excluded from the anonymous evidence package.

Scene	Method	Collision (%)	Success (%)	Intervention step fraction (%)	Recovery/episode	Success time (s)
Single	ABER	0	100	49.73	8.57	6.440
Single	ABER-FP	0	100	23.33	0	5.265
Multi	ABER	0	100	83.01	13.61	7.308
Multi	ABER-FP	0	100	34.49	0	6.286

Table 9: Frozen held-out comparison in the exact differential-drive model. Each method/scene cell uses 500 non-development seeds. All 2,000 runs have zero certificate violations.

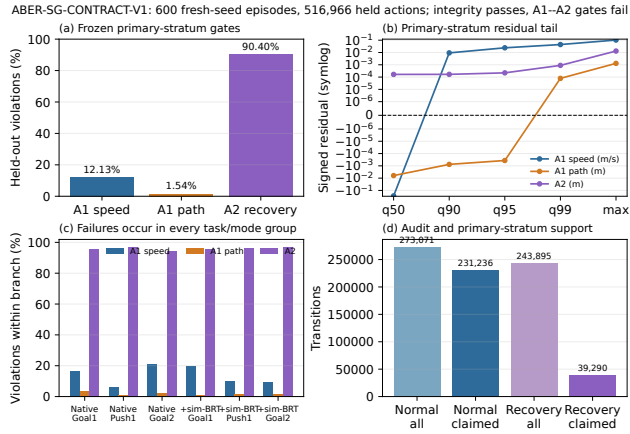


Figure S2: The held-out contract audit. Failures remain after excluding every hold with non-ground contact and occur in every task/training-mode group.

Frozen ABER-FP: 500 held-out episodes per method/scenario; zero collisions and certificate violations

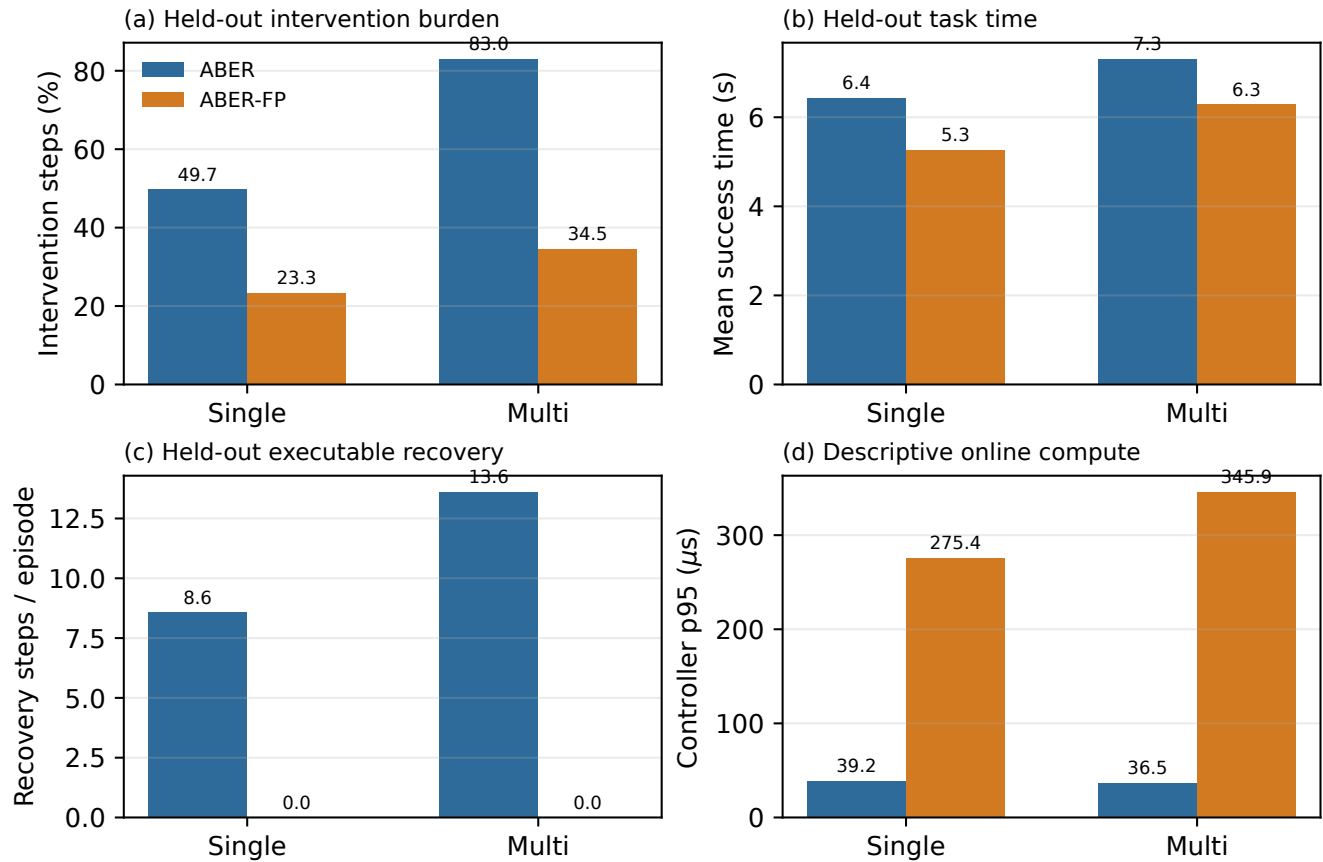


Figure S3: Held-out task and intervention performance, with descriptive controller latency. ABER-FP lowers intervention occupancy by 53.08% in the single-obstacle scene and 58.45% in the multi-obstacle scene, while reducing mean success time by 18.24% and 13.98%, respectively.